For this study, we have curated a set of six datasets that are well-suited for assessing knowledge editing methods. A detailed statistical overview of these datasets is presented in Table 3, and they encompass a range of editing types, including fact manipulation, sentiment modification, and hallucination generation.

Focusing on the task of knowledge insertion, we have adopted the dataset, WikiData$_{recent}$ [157]:

- **WikiData**$_{recent}$ This dataset specifically focuses on triplets that have been recently inserted into WIKIDATA after July 2022. Consequently, this dataset enables us to create insertion edit requests for models that were trained prior to the introduction of these facts, thereby simulating scenarios where an outdated model meets the new world knowledge. We utilize the original datasets provided by the authors and split them into training and testing sets.

For knowledge modification, we have selected the following four datasets: ZsRE [185], WikiBio [163], Wikidata$_{recent}$ [157], and Convsent [153].

- **ZsRE** is a context-free question-answering task. Given a question based on the subject and relation, the model is expected to provide the correct object as the answer. We adopt the extended version of ZsRE proposed by Yao et al. [69], which introduces a portability test for the original dataset. Additionally, we collect new locality sets following the procedure outlined in Yao et al. [69], as the original dataset computes locality using Natural Question annotations.
- **WikiBio** The original dataset was created by prompting GPT-3 to generate 238 Wikipedia-style biographies using subjects from the WikiBio dataset [186]. Hartvigsen et al. [163] utilizes this dataset and introduces a new editing task focused on correcting hallucinations in GPT language models. They annotate the factual accuracy of each sentence, identifying the ones that contain hallucinations. We follow their approach by editing inaccurate sentences and replacing them with corresponding sentences from the true Wikipedia entries. We adhere to the original setting of this dataset and construct the locality set by linking concepts via the Wikidata API to traverse all relations of the concept and randomly select an unrelated relationship and tail entity.
- **WikiData**$_{counterfact}$ Since tail entities are often not captured by models, and therefore are not suitable for testing modification edits [187], [157] collect triplets about popular entities, where the subject corresponds to one of the top-viewed pages in Wikipedia. They also collect a dataset by random sampling entities from Wikidata, and we use it as the training set and the WikiData$_{counterfact}$ as the test set.
- **ConvSent** is a sentiment editing task that assesses the model's ability to modify a dialog agent's sentiment on a specific topic without affecting its responses to other topics. For example, given the topic 'What do you think of bananas?', we wish the post-edited model to give the corresponding sentiment for 'bananas' including positive and negative. The locality sets consist of examples generated from entities other than the one used for editing. We also adopt the original setting of the ConvSent dataset.

In the context of knowledge erasure settings, we have selected the Sanitation [188] dataset.

- **Sanitation** This dataset specifically addresses privacy concerns associated with learned language models. It focuses on the task of forgetting specific information stored in the model. The dataset provides pairs of questions and answers, where the answers contain knowledge that needs to be forgotten (e.g., "1234 Oak Street"), and the questions prompt the model to generate the corresponding answers (e.g., "What is John Smith's address?"). The goal is for the post-edited model to effectively forget the target answer and generate predefined safe token sequences, such as "I don't know," in response to prompts seeking specific or sensitive information. This mechanism helps prevent information leakage. The dataset consists of a forgot set and a retain set. We utilize the forget set to evaluate the success of the model's editing process and the retain set to assess the locality of the modifications. Furthermore, we maintain the original task settings by sampling the same number of data instances as the training set.

In addition to the datasets we have selected, the literature offers a diverse range of knowledge editing tasks, each addressing specific aspects and challenges in this domain. **DepEdit** [189] is a more robust analysis dataset that delves into the internal logical constraints of knowledge, offering a deeper

understanding of knowledge structures. Notably, Xu et al. [190] introduces cross-lingual model editing tasks and further proposes language anisotropic editing to improve cross-lingual editing by amplifying different subsets of parameters for each language. In the case of multilingual models, changes in one language within multilingual models should result in corresponding alterations in other languages. **Eval-KLLM** [164] and **Bi-ZsRE** [191] have been designed to assess the cross-lingual editing capabilities of models. Wang et al. [192] proposed Retrieval-augmented Multilingual Knowledge Editor (ReMaKE), which is capable of performing model-agnostic knowledge editing in multilingual settings. The authors also offer a multilingual knowledge editing dataset (**MzsRE**) comprising 12 languages. Another dataset, **ENTITY INFERENCES** [193], focuses on entity propagation, where the model is provided with a definition and asked to reason based on the given definition. Time-series knowledge editing is explored in **TEMPLAMA** [156] and **ATOKE** [194], where the objective is to modify knowledge pertinent to specific time periods without affecting other temporal knowledge. For commonsense knowledge editing, Gupta et al. [152] introduced $\textbf{MEMIT}_{\textbf{CSK}}$, applying existing editing techniques to modify commonsense knowledge within models. Furthermore, **RaKE** [195] is proposed to measure how current editing methods edit relation knowledge. All previous work usually confines the edit as a knowledge triplet. Akyürek et al. [196] proposes a new dataset **DUNE** that broadens the scope of the editing problem to include an array of editing cases, such as debiasing and rectifying reasoning errors, and defines an edit as any natural language.

It is important to note that some of these datasets may be just published or not currently available. Therefore, in this paper, we focus on evaluating the performance and effectiveness of knowledge editing techniques within some popular works. We plan to expand our benchmark in the future as we acquire new datasets. For additional related datasets, please refer to Wang et al. [70].

3.5 Evaluation for Knowledge Editing

Knowledge editing aims to alter model behavior based on modified facts. However, knowledge is interconnected; changing one fact may ripple outwards and affect other facts in complex ways. This interdependence makes assessing the effects of editing difficult. We summarize key evaluation criteria from prior work into four categories: edit success, portability, locality, and fluency.

Edit Success The purpose of editing is to change the model's output of given knowledge. Previous work adopt two metrics named ***reliability*** and ***generalization***. In reliability testing, the goal is to evaluate whether the post-edited model can provide the target answer for a given context. On the other hand, generalization testing aims to assess the post-edited model's performance on paraphrased contexts. However, for knowledge editing tasks, the primary objective is to modify the underlying factual knowledge rather than just altering its expression. Consequently, both the given text and its paraphrased versions should undergo changes to reflect the edited knowledge. Here, we follow previous work [170, 172] and collectively refer to reliability and generalization the as **edit success**. Hence, here, edit suceess means the post-edit model should not only answer the question itself correctly but also give the right answer for input with similar expressions.

Portability Meanwhile, knowledge is not isolated, and solely changing the given knowledge is not enough for downstream use. When the knowledge is corrected, the model is supposed to reason about the downstream effects of the correction. Here, we follow previous work [157, 69, 155] to evaluate whether the edited model can address the implications of an edit for real-world applications and name it as portability to evaluate what would ensue after the knowledge editing. Portability contains three different parts:

- **Alias:** The editing of one subject should not vary from its expression. Wikidata maintains a set of aliases for every entity. Hence, here, we follow Cohen et al. [157], Yao et al. [69] to replace the question's subject with an alias or synonym to evaluate post-edited model's performance on other descriptions of the subject.
- **Compositionality and Reasoning:** This requires the post-edit model to conduct reasoning with the changed facts. For example, when we change the current president of the U.S. from Donald Trump to Joe Biden, the answer to the question "Who is the First Lady of the United States?" should also be changed.
- **Logical Generalization:** These are the changes that are semantically related to the modified fact and expected to change by the edit; they were indeed modified. For example, as

mentioned by Yao et al. [69], when the fact of (s, r, o) are changed, the reversed relation of the knowledge $(o, \hat{r}, s)$ should also be changed.

Locality When editing the knowledge, we may inadvertently change the knowledge that we don't want to modify. A good edit is supposed to modify the knowledge locality without influencing the knowledge that is unrelated. The evaluation of locality includes two levels:

- **In-Distribution**: this one includes the knowledge that comes from the same distribution. As shown in previous work, overediting is a common phenomenon. Here, we follow Meng et al. [79], Cohen et al. [157], Yao et al. [69] and construct the related in-distribution knowledge, including *forgetfulness* and *relation specificity*. Forgetfulness evaluates whether the post-edit model retains the original objects in one-to-many relationships. The principle of relation specificity posits that any other attributes of the subject, which have been previously updated, should remain unaltered following the editing process.
- **Out-of-Distribution**: the other knowledge that is not associated with the target one should not be influenced. That is, we also don't want the edited model to lose their general ability to deal with other tasks. Hence, here we test the edited model on the popular NLP benchmark in Section 4.2.

It should be noted that some work use **Specificity** to denote locality.

Generative Capacity Previous work find that, after editing the model, some models tend to generate repeated things and often generate the edited target whenever encountering the subject words. Additionally, the metric **fluency** are employed to evaluate the generative capacity of the post-edited model. Here we follow ROME [79] and employ the *fluency* to measure the model's generation ability after editing. In particular, we calculate the weighted average of bi-gram and tri-gram entropies to assess the diversity of text generations. A decrease in this value indicates increased repetitiveness in the generated text.

4 Experiments

In our study, we conduct experiments using current methods and datasets to investigate knowledge editing techniques in the context of LLMs. By conducting experiments using these methods and leveraging appropriate datasets, we aimed to evaluate the performance and efficacy of knowledge editing techniques in LLMs. Our goal was to gain insights into the challenges, limitations, and potential improvements associated with editing knowledge in these models.

4.1 Experiment Settings

We choose **Llama2-7b-chat** [197] as our base model, specifically its chat version, which has demonstrated improved consistency after reinforcement learning from human feedback (RLHF). The model generates an answer to each question with greedy autoregressive decoding. To establish baselines for comparison, we employed eight model editing methods that have shown effectiveness in prior research. These methods were selected based on their ability to modify the knowledge within LLMs [69]. As a further baseline strategy, we also used the fine-tuning method (**FT-L**) put forth by Meng et al. [79]. **FT-L** directly fine-tunes a single layer's feed-forward network (FFN), specifically the layer identified by the causal tracing results in ROME. This method uses the last token's prediction to maximize the probability of all tokens in the target sequence immediately, deviating from the original fine-tuning objective. To address this, we also experiment with an improved fine-tuning method, **FT-M**. It trains the same FFN layer as FT-L using the cross-entropy loss on the target answer while masking the original text. This approach aligns more closely with the traditional fine-tuning objective. For the in-context learning methods, we use the ICE method proposed by Cohen et al. [157]. This method prepends a prompt 'Imagine that {knowledge}' before the input.

All the experiments are conducted by EasyEdit [198]. As to the evaluation of the post-edited model, some of the previous works computed the probability difference of the output for pre-edit and post-edit models: $P[y^*|\theta^{'}] - P[y|\theta]$. y^* is the edit target, and y is the original model's prediction. However, the higher probability for y^* does not mean an idea outcome, and for realistic usage, when we edit the model, we hope it generates the desired output. Hence, for the evaluation of fact datasets